

1. Administrative & Study Identification

Full Study Title: A Phase I/II Open Label Study to Evaluate the Safety, Cellular Kinetics, and Efficacy of AZD0754, a Chimeric Antigen Receptor (CAR) T-cell Therapy Directed Against STEAP2, in Adult Participants With Metastatic Prostate Cancer: APOLLO **Short Title/Acronym:** APOLLO **Protocol Number:** D9660C00001 **NCT Number:** NCT06267729 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD0754 (STEAP2-directed autologous CAR T-cell therapy) **Phase of Development:** Phase I/II **Medical Monitor:** AstraZeneca Clinical Operations Lead / Study Physician

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety, tolerability, and antitumour activity of AZD0754 CAR T-cell therapy in participants with metastatic prostate cancer, including determining Dose Limiting Toxicities (DLTs) and the recommended dose for the expansion phase. **Secondary Objectives:** To assess the cellular kinetics, pharmacodynamics, and preliminary efficacy, including PSA-related changes (PSA50, PSA90).

2.2 Background & Rationale To evaluate a novel targeted CAR T-cell therapy (AZD0754) directed against STEAP2 in adult male patients with metastatic castration-resistant prostate cancer (mCRPC) who have progressed on previous standard therapies (including novel hormonal agents [NHAs] and taxanes) and have limited treatment options.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Monotherapy) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** 60 participants (Estimated) **Number of Sites:** Multi-center **Estimated Study Start:** 12-Mar-2024 **Estimated Primary Completion:** 24-Nov-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants, age \$\\ge\$ 18\$ years, weighing more than 39 kg. 2. Histologically confirmed diagnosis of metastatic adenocarcinoma of the prostate without known neuroendocrine differentiation or small cell features. 3.

Castration-resistant prostate cancer (mCRPC) with measurable PSA ≥ 1 ng/mL and evidence of radiographic progression in soft tissue or bone within 6 months prior to screening.

4.2 Exclusion Criteria 1. History of another primary malignancy (except for non-melanoma skin cancer or those treated with curative intent ≥ 2 years ago) or known brain metastases. 2. Uncontrolled intercurrent illness including active infection, symptomatic congestive heart failure (NYHA class ≥ 3), interstitial lung disease, or myocardial infarction within the past 6 months. 3. Active, uncontrolled epilepsy or history of stroke, intracranial hemorrhage, or seizure within 6 months of apheresis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Autologous CAR T-cells (AZD0754) administered as a single intravenous (IV) infusion. **Route of Administration:** Intravenous. **Duration of Treatment:** Single infusion followed by a protocol-specified in-hospital monitoring period.

5.2 Concomitant & Prohibited Therapy Permitted: Leukapheresis, bridging therapy for disease control during CAR-T production, protocol-defined lymphodepleting chemotherapy (LDC - cyclophosphamide, fludarabine, or bendamustine), and ongoing medical castration therapy. **Prohibited:** Any concurrent anticancer treatment not specified in the protocol; Radiotherapy within 14 days of apheresis (unless the radiation portal covered $\leq 5\%$ of the bone marrow reserve).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Pre-treatment	Informed Consent, Medical History, Leukapheresis, Bridging Therapy (if needed)
Baseline	Pre-infusion to Day 0	Lymphodepleting Chemotherapy (LDC), AZD0754 IV Infusion
Interim	Post-infusion	In-hospital monitoring, Safety Labs, Cellular Kinetics, PSA Assessment
End of Study	Up to Nov 2026	Final Vital Signs, Exit Interview, Long-term Safety/Efficacy Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs) and Adverse Events (AEs, AESIs, SAEs) to establish safety and determine the recommended dose for the Phase II expansion phase. **Measurement Method:** Clinical observation, safety monitoring, and laboratory assessments (e.g., CTCAE grading).

7.2 Secondary & Exploratory Endpoints Prostate-Specific Antigen (PSA) related changes (e.g., PSA50, PSA90). Cellular kinetics and pharmacodynamics of AZD0754.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring post-infusion during the required hospital stay; collection of AEs and AESIs. **Serious Adverse Events (SAEs):** Routine recording and expedited reporting per standard sponsor timelines. **Data Monitoring Committee (DMC):** AstraZeneca internal and external committee review for Phase I dose-escalation clearance.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive any study intervention) and Efficacy Set. **Sample Size Justification:** Target $N=60$ configured to support Phase I safety/dose-finding and Phase II preliminary efficacy requirements. **Handling of Missing Data:** Standard handling aligned with early-phase oncology guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict onsite monitoring during LDC, cell infusion, and post-infusion recovery periods. **Audit Trail:** Electronic Case Report Form (eCRF) locking and robust Chain of Identity/Chain of Custody (COI/COC) tracking for autologous cell product manufacturing.

11. Study Identifiers & Governance

Primary ID: D9660C00001 **Registry Number:** NCT06267729 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** APOLLO **Official Regulatory Title:** A Phase I/II Open Label Study to Evaluate the Safety, Cellular Kinetics, and Efficacy of AZD0754, a Chimeric Antigen Receptor (CAR) T-cell Therapy Directed Against STEAP2, in Adult Participants With Metastatic Prostate Cancer: APOLLO

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Progression despite castration therapy, PSA ≥ 1 ng/mL, prior NHA and taxane exposure **Medical Methodology:** STEAP2-directed Autologous CAR T-

cell therapy **Optimization Target:** Maximum Tolerated Dose (MTD) / Recommended Phase II Dose

13. Study Procedures & Methodology

Management Pathway: Leukapheresis → Bridging Therapy → Lymphodepleting Chemotherapy (LDC) → AZD0754 Infusion **Education Protocol (HCP):** Site training on CAR-T administration and specialized toxicity management (e.g., Cytokine Release Syndrome). **Education Protocol (Patient):** Informed consent surrounding apheresis, intensive in-patient monitoring, and recognition of CAR-T related side effects. **Quality Audit Mechanism:** Tracked apheresis cellular extraction, shipping, manufacturing feasibility, and product administration.

14. Observation & Follow-Up Schedule

Total Duration: Estimated 32 months (March 2024 to November 2026) **Onsite Visit Cadence:** Intensive monitoring at the clinical site required during the protocol-specified post-infusion period. **Remote Monitoring Frequency:** Routine safety and PSA/PK check-ins post-hospital discharge. **Checklist Review Interval:** Regular oncological intervals corresponding to disease progression staging.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				